regime	$\partial_t \mathbf{B}$ Faraday's law B → E	$\partial_t \mathbf{D}$ Ampère's law E → B	$\sigma \mathbf{E} \rightarrow \mathbf{B}$ induced current eddy feedback	$\partial_t \rho$ charge build-up enters $\nabla \cdot \mathbf{J}$	typical problems
Static $\omega = 0$	no time change	no time change	no time change	no time change	DC conduction, electro-statics, magnets
EQS $\eta_m \ll 1$	dropped	**kept**	dropped	**kept** drop if $\eta_d \ll 1$	TES, electrodes, capacitive coupling
MQS $\eta_w \ll 1$	**kept**	dropped	**kept**	dropped	eddy currents, shields, gradient coil + cryostat
One-way MQS $\eta_m \ll 1,\ \mu \approx \mu_0$	**kept**	dropped	dropped	dropped	TMS; gradient fields in the body
Full wave none	**kept**	**kept**	**kept**	**kept**	MRI RF, SAR